BMS COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF MATHEMATICS

Sem & Branch:	**4th Semester CS, CS-DS, AI-DS, CS-IOT**	**Subject:**	**Linear Algebra and Optimization**	**Course Code:**	**23MA4BSLAO**
Time:	**12:00 PM to 1:15 PM**	**Test Date:**	**06/04/2026**	**Max Marks: 40**	

Test - 1

Test No.	Q. No.	**Scheme of Valuation**	Marks
	1	In a study to understand the relationship between daily practice time and performance improvement, a researcher records how many hours per day a person practices a certain skill (x) and how much their test score improves after one week (y)	6

x (hours/day)	0.5	1.0	1.5	2.0	2.5	3.0
y (score gain)	2.1	4.0	4.8	5.9	6.7	8.0

Determine the least squares straight line and hence predict the improvement in scores, if a person practices for 5 hours per day.

Ans

$A=\begin{bmatrix}1 & X\end{bmatrix}$, $B=\begin{bmatrix}Y\end{bmatrix}$ with $AX=B \Rightarrow A^T AX = A^T B$ — 1+1

$A=\begin{bmatrix}6 & 10.5\\ 10.5 & 22.75\end{bmatrix}$ $B=\begin{bmatrix}31.5\\ 64.8\end{bmatrix}$ — 2

$y=1.38095+2.21143x$ — 1

$y(5)=12.44$ — 1

2 Apply Cayley-Hamilton theorem to compute A^4 given $A=\begin{bmatrix}2 & 3 & -1\\ 2 & -1 & 2\\ 1 & 2 & 3\end{bmatrix}$. Use minimal matrix multiplication. — 6

Ans

$-\lambda^3+4\lambda^2+8\lambda-31=0$ — 2

$A^3-4A^2-8A+31I=0$ — 1

$A^3=4A^2+8A-31I$ — 1

$A^4=24A^2+A-124I$ — 1

$A^4=\begin{bmatrix}94 & 27 & 23\\ 98 & 139 & 50\\ 217 & 170 & 167\end{bmatrix}$ — 1

3 It is known that $\mathbb{R}^2$ is a vector space over the field of reals. Consider the mapping $\langle\ ,\ \rangle:\mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R}^2$ defined by $\langle u,v\rangle=2u_1v_1-u_1v_2+u_2v_1+u_2v_2$. Verify all the properties of the inner product space. Is $(\mathbb{R}^2,\langle\ ,\ \rangle)$ an inner product space? — 7

Ans

Positive definiteness: $\langle u,u\rangle=2u_1^2+u_2^2 \Rightarrow \langle u,u\rangle\geq 0$

$\langle u,u\rangle=0 \Rightarrow u_1=0=u_2 \Rightarrow u=0$

Mapping satisfies positive definite property. — 2

$\langle u,v\rangle=2u_1v_1-u_1v_2+u_2v_1+u_2v_2$ and $\langle v,u\rangle=2v_1u_1-v_1u_2+v_2u_1+v_2u_2$ — 1

$\Rightarrow \langle u,v\rangle\neq\langle v,u\rangle$. Hence the mapping **does not** satisfy symmetric property. — 1

Linearity: $\langle au+bv,w\rangle=a\langle u,w\rangle+b\langle v,w\rangle$

$\langle a\mathbf{u}+b\mathbf{w},\mathbf{v}\rangle=2(au_1+bw_1)v_1-(au_1+bw_1)v_2+(au_2+bw_2)v_1+(au_2+bw_2)$

		$= a(2u_1v_1 - u_1v_2 + u_2v_1 + u_2v_2) + b(2w_1v_1 - w_1v_2 + w_2v_1 + w_2v_2)$ $\langle au+bv,w\rangle = a\langle u,w\rangle + b\langle v,w\rangle \Rightarrow$ The given mapping satisfies linearity property. Alternate to linearity, one can prove $[\langle u+v,w\rangle = \langle u,w\rangle + \langle v,w\rangle$ *and* $\langle au,v\rangle = a\langle u,v\rangle]$. The mapping is not an inner product on $\mathbb{R}^2$.	2 1
	4	Find the orthogonal complement of the subspace spanned by $S=\{u_1,u_2\}$ where $u_1 = 2t+1$ and $u_2 = 2t-1$ in the polynomial space $P_2(t)$ with the inner product $\langle f(t),g(t)\rangle = \int_{-1}^{1} f(t)g(t)dt$.	7
	Ans	Let $v = at^2+bt+c$. Given $\langle v,u_1\rangle = 0$ and $\langle v,u_2\rangle = 0$ $\langle v,u_1\rangle = 0 \Rightarrow \int_{-1}^{1} v.u_1 dt = \int_{-1}^{1}(at^2+bt+c)(2t+1)dt = 0$ $\int_{-1}^{1}[2at^3+(a+2b)t^2+(b+2c)t+c]dt = 0 \Rightarrow \frac{4b}{3}+\frac{2a}{3}+2c=0$ $\langle v,u_2\rangle = \int_{-1}^{1} v.u_2 dt = \int_{-1}^{1}[at^3+(2b-a)t^2+(2c-b)t-c]dt = 0 \Rightarrow \frac{4b}{3}-\frac{2a}{3}-2c=0$ $\Rightarrow b=0, a=-3k, c=k \Rightarrow v = k(-3t^2+1)$.	1 2 2 1+1
	5	Show that $S=\{(1,0,1,1),(-1,1,1,0),(-1,-1,0,1),(0,1,-1,1)\}$ is an orthogonal basis of $\mathbb{R}^4$. Hence compute the coordinate vector of $(1,-16,5,0)$ relative to the basis S.	7
	Ans	$\langle u_1,u_2\rangle = 0$, $\langle u_1,u_3\rangle = 0$, $\langle u_1,u_4\rangle = 0$, $\langle u_2,u_3\rangle = 0$, $\langle u_2,u_4\rangle = 0$ and $\langle u_3,u_4\rangle = 0$ S is an orthogonal set and hence it is a linearly independent set. Any independent set of n vectors in an n - dimensional vector space V is a basis V Hence S is an orthogonal basis of $\mathbb{R}^4$ $[v]_s = c_1u_1+c_2u_2+c_3u_3+c_4u_4 = \frac{6}{3}u_1+\frac{-12}{3}u_2+\frac{15}{3}u_3+\frac{-21}{3}u_4$	2 1 1 3
	6a	Obtain the orthogonal basis of the rows of the matrix $\begin{bmatrix}1&2&1&1\\2&0&3&1\\1&-1&1&0\end{bmatrix}$.	7
	Ans	$u_1=(1,2,1,1)$, $u_2=(2,0,3,1)$ and $u_3=(1,-1,1,0)$ $v_1=u_1=(1,2,1,1)$ $v_2=(2,0,3,1)-\frac{6}{7}(1,2,1,1)=\left(\frac{8}{7},\frac{-12}{7},\frac{15}{7},\frac{1}{7}\right)$. Let $v_2=(8,-12,15,1)$ $v_3=(1,-1,1,0)-\frac{0}{7}(1,2,1,1)-\frac{35}{434}(8,-12,15,1)=\left(\frac{11}{31},\frac{-1}{31},\frac{-13}{62},\frac{-5}{62}\right)$ $v_3=(22,-2,-13,-5)$	1 2 2 3
	OR		

	6b	Consider a subspace of $\mathbb{R}^5$ spanned the vectors u_1 and u_2 where $u_1=(2,2,-3,4,0)$ and $u_2=(3,1,0,-2,1)$. (i) Find the angle between u_1 and u_2. (ii) Find the projection of $v=(1,2,1,2,1)$ along W.	7
	Ans	$\langle u_1,u_2\rangle=6+2+0-8+0=0 \Rightarrow u_1$ and u_2 are orthogonal. $\text{Proj}(v,W)=c_1u_1+c_2u_2$ $c_1=\frac{11}{33}$ and $c_2=\frac{2}{15}$ $\text{Proj}(v,W)=\frac{1}{3}(2,2,-3,4,0)+\frac{2}{15}(3,1,0,-2,1)=\left(\frac{16}{15},\frac{12}{15},-1,\frac{16}{15},\frac{2}{15}\right)$	1 1 4 1

Appropriate marks may be awarded to alternate methods

Solution and scheme prepared by:

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